

# MATHS: STATISTICS — BOX PLOTS & IQR



## What is Statistics?

Statistics helps us collect, organize, and understand data!

- Count favorite fruits in your class
- Measure heights of friends
- Track daily temperatures

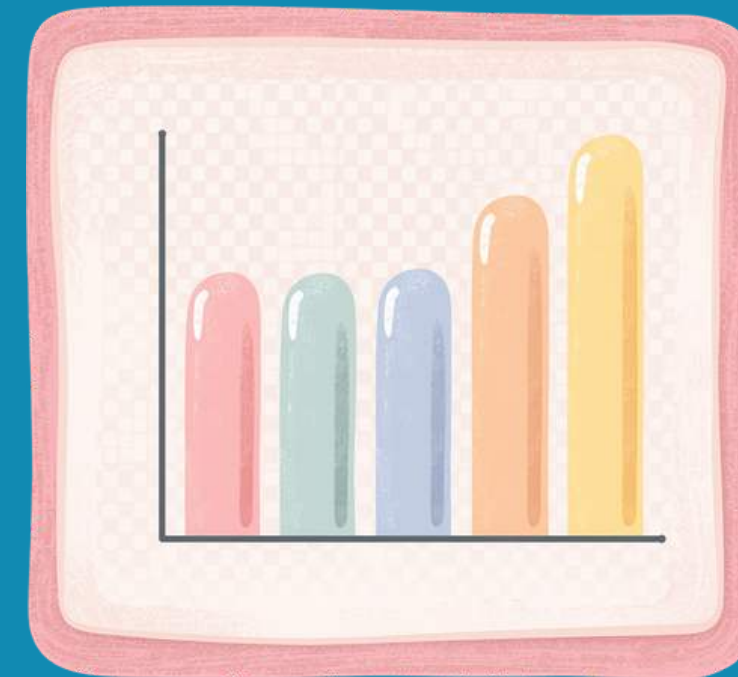
## Why is it useful?

- Helps us see patterns
- Makes decisions easier
- Answers questions with facts



A data set is a collection of numbers or information.

Distribution shows how data values spread out or cluster together.





The smallest  
number in  
your data set



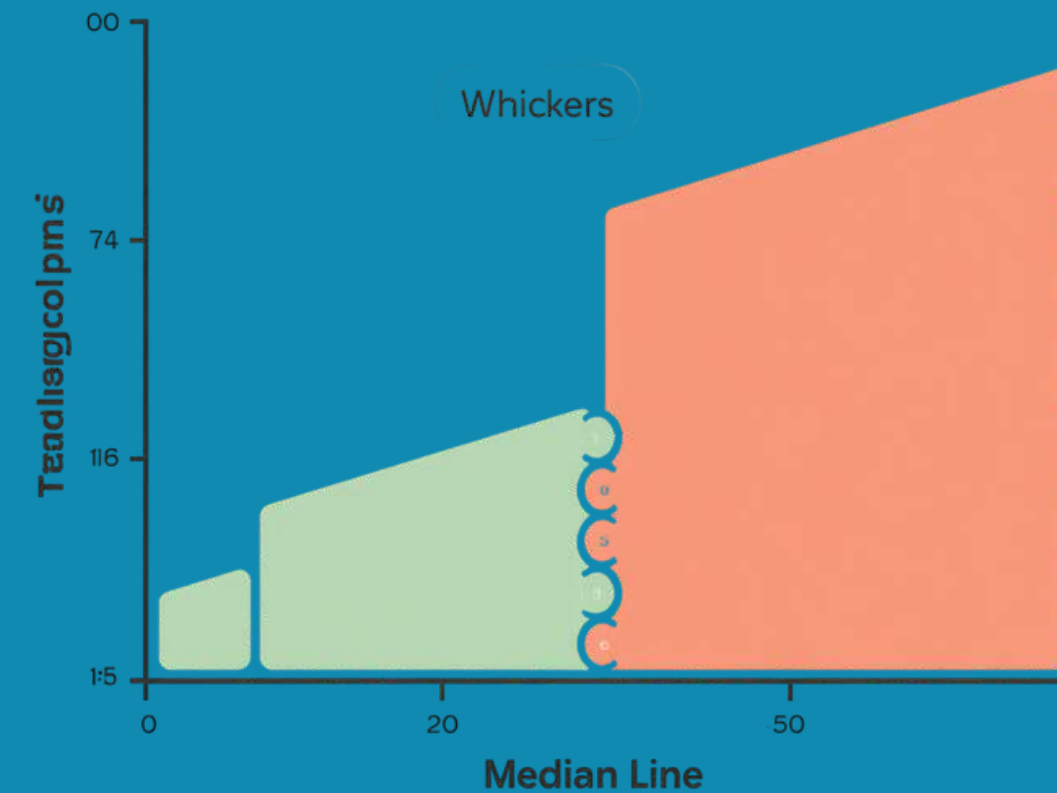
The middle  
number when  
data is ordered



Maximum is the  
largest number  
in your data set

A box plot shows the five-number summary visually, helping us see data spread quickly.

### Box plot diagrams







1

2

## STEP 1

Order your data  
from smallest to  
largest number

## STEP 2

Find the median  
(middle number  
of your data)

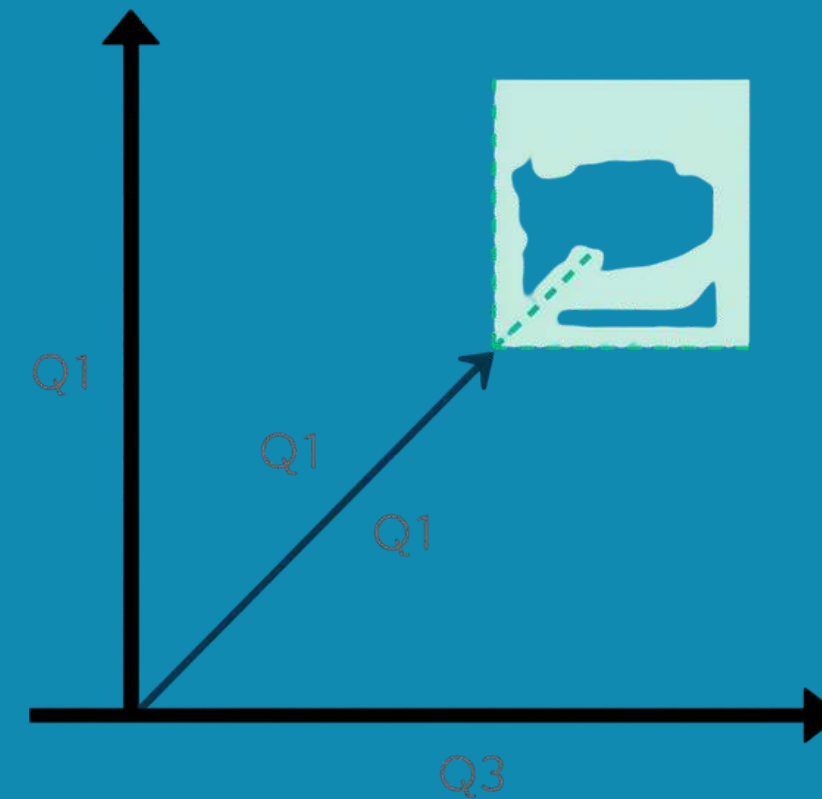
## STEP 3

Find Q1 & Q3  
(medians of  
lower and upper  
halves)



$$\text{IQR} = Q3 - Q1$$

The Interquartile Range shows the middle 50% of your data. It helps us measure spread and spot outliers!





Find the five-number summary: Min, Q1, Median, Q3, Max



Draw a number line covering your data range



Draw box from Q1 to Q3 with median line, add whiskers to min & max



## Reading Box Plots

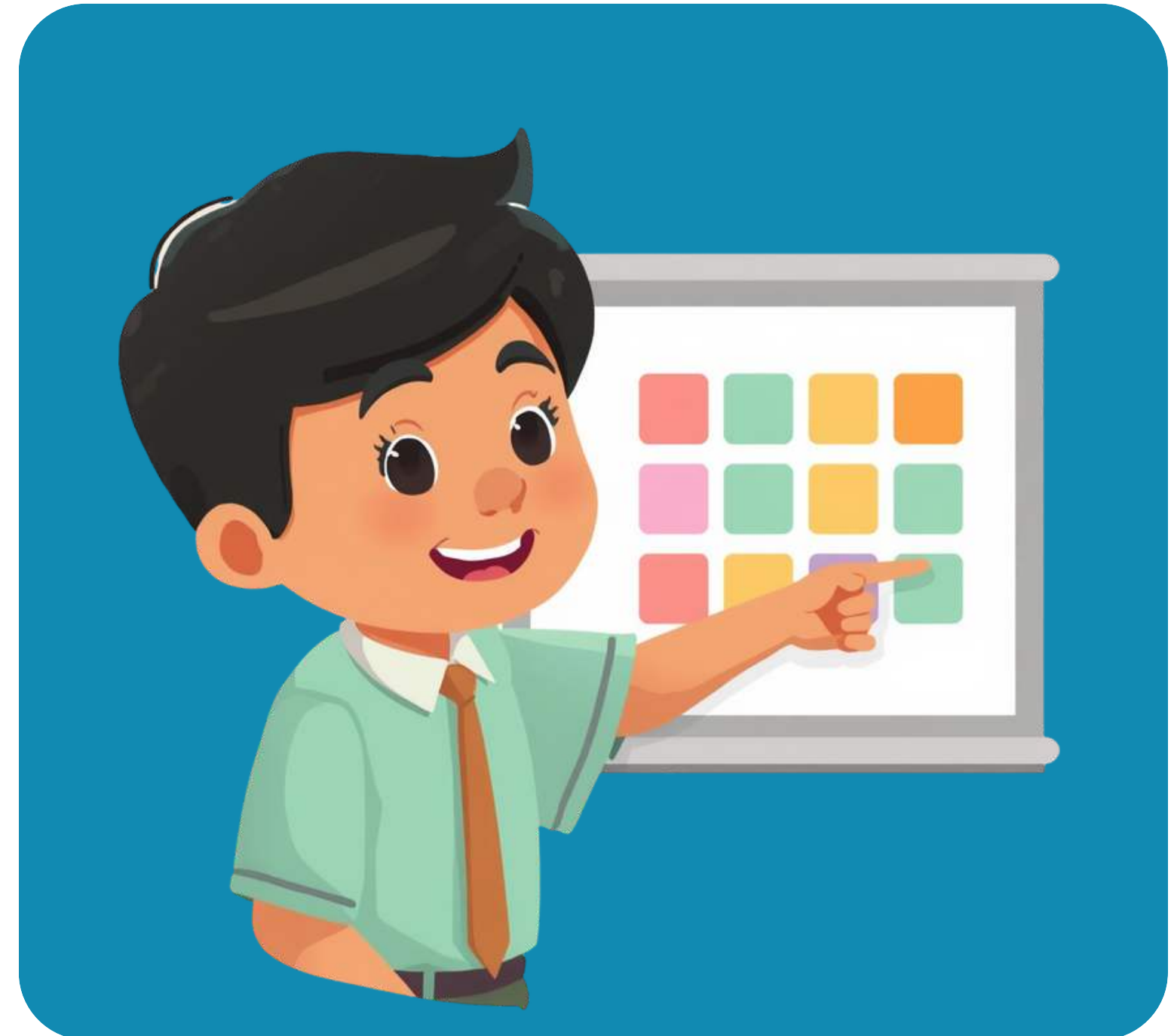
The box shows the middle 50% of data (from Q1 to Q3).

The line inside the box marks the median.

Whiskers extend to the minimum and maximum values.

Symmetrical box plots have equal whiskers and a centered median.

Skewed plots have longer whiskers on one side!



Box plots help us  
compare heights of  
classmates, track  
daily temperatures,  
or see the spread of  
shoe sizes in class!



Comparing two box plots helps us see differences in data! Look at the medians to compare typical values, and check the IQR to see which group has more spread.



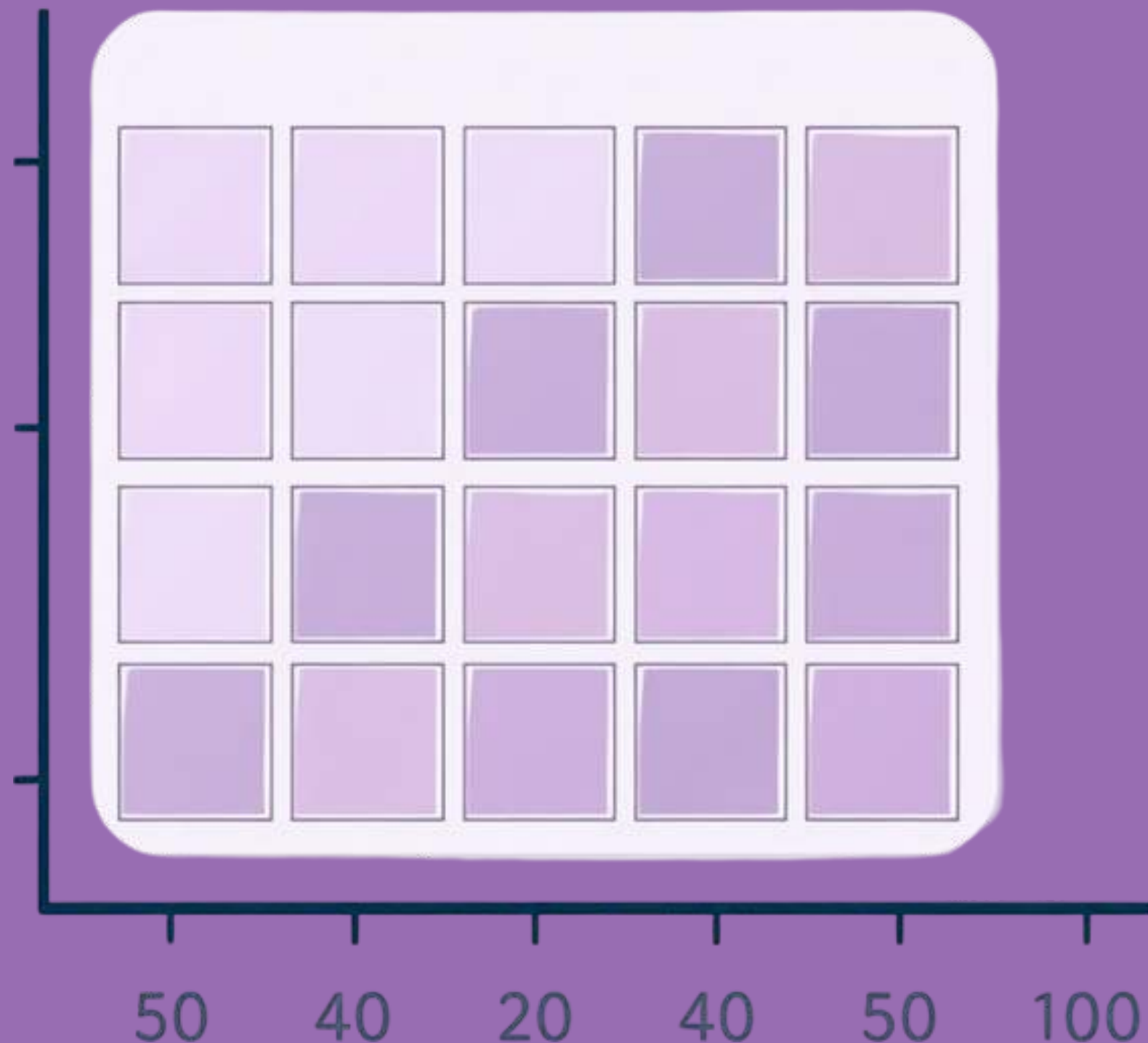
# PRACTICE PROBLEM 1: CAN YOU FIND THE IQR?



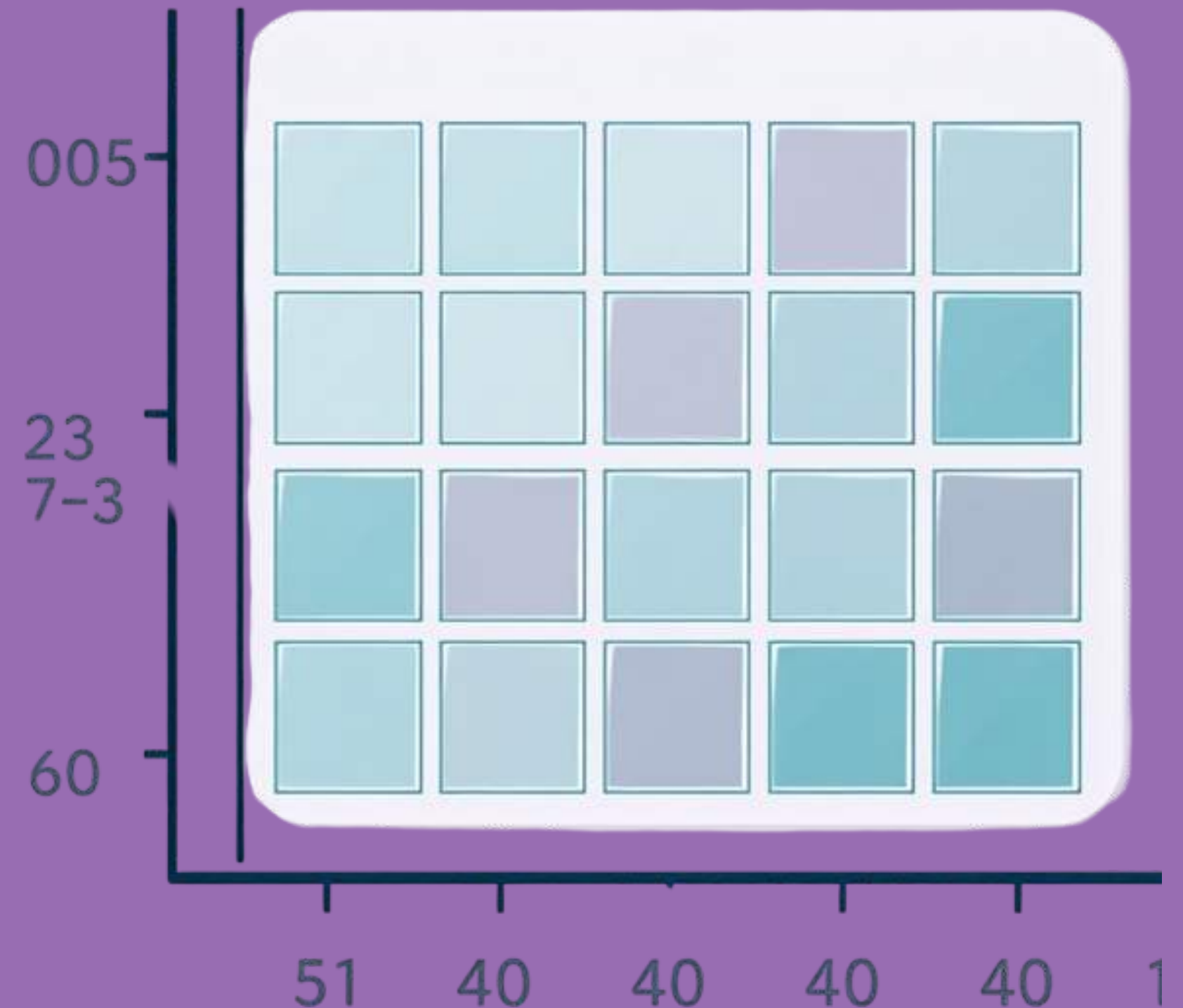
|    |    |    |    |
|----|----|----|----|
| 1  | 15 | 20 | 20 |
| 21 | 23 | 28 | 28 |

# PRACTICE PROBLEM 2: READING BOX PLOTS

Class A



Class B





## FIVE-NUMBER SUMMARY



5 —  
4 —  
5 —

Includes minimum,  
Q1, median, Q3, and  
maximum values

## BOX PLOTS



Visualize data  
spread using the  
five-number  
summary

## IQR



IQR

IQR measures  
the middle 50%  
of data spread



# KEEP EXPLORING STATISTICS!



THANK YOU &

QUESTIONS?